

DAY 7 – QUESTION

Basic Test Function Write a pytest test function that checks whether the sum of two numbers (3 and 5) equals 8.

Assertion Failure Create a pytest test that intentionally fails by asserting that "hello".upper() equals "hello".

Fixture Usage Define a pytest fixture that returns a list of numbers [1, 2, 3]. Write a test that uses this fixture to verify the list length is 3.

Parameterized Test Use @pytest.mark.parametrize to test a function square(x) for inputs 2, 3, 4 and expected outputs 4, 9, 16. **Exception Handling**

Write a pytest test that verifies a ZeroDivisionError is raised when dividing by zero using pytest.raises.

CODE

```
import pytest
def test_sum():
    assert 3 + 5 == 8
def test_upper_fail():
    assert "hello".upper() == "hello"
@pytest.fixture
def sample_list():
    return [1, 2, 3]
def test_list_length(sample_list):
    assert len(sample_list) == 3
def square(x):
    return x * x
@pytest.mark.parametrize("input, expected",
                          [(2, 4), (3, 9), (4, 16)])
def test_square(input, expected):
    assert square(input) == expected
def divide(a, b):
    return a / b
def test_divide_by_zero():
    with pytest.raises(ZeroDivisionError):
        divide(10, 0)
```

OUTPUT

collected 7 items

```
test_assignmentpytest.py::test_sum PASSED
[ 14%]
test_assignmentpytest.py::test_upper_fail FAILED
[ 28%]
test_assignmentpytest.py::test_list_length PASSED
[ 42%]
test_assignmentpytest.py::test_square[2-4] PASSED
[ 57%]
test_assignmentpytest.py::test_square[3-9] PASSED
[ 71%]
test_assignmentpytest.py::test_square[4-16] PASSED
[ 85%]
test_assignmentpytest.py::test_divide_by_zero PASSED
[100%]
```

FAILURES

```
=====
test_upper_fail
=====
```

```
def test_upper_fail():
>     assert "hello".upper() == "hello"
E     AssertionError: assert 'HELLO' == 'hello'
E
E     - hello
E     + HELLO
```

```
test_assignmentpytest.py:7: AssertionError
```

```
===== short test
```

```
summary info =====
```

```
FAILED test_assignmentpytest.py::test_upper_fail - AssertionError: assert 'HELLO' == 'hello'
```

```
===== 1 failed, 6 passed in
```

```
0.16s =====
```

```
(.venv) PS C:\WiproTraining\pytestassign\pytestassignment>
```